

Banking Data Analysis

Customer, Account, Transaction & Loan Analytics



Prepared By : Vaddi Pavan Aditya



Tools Used: Excel | SQL | Python | Power BI

Project Description:

This project analyzes customer, account, loan, and transaction data using SQL, Python, Excel, and Power BI to generate actionable business insights, identify performance trends, and support data-driven decision-making for banking operations.

Dataset Overview

- 5,000 Customers
- 5,000 Active Accounts
- 100,000+ Banking Transactions
- ₹603.12 Million Total Transaction Value
- ₹6.0 Billion Loan Portfolio
- ₹60K Average Account Balance
- Reporting Period: 2022–2025

Project Objective

The objective of this project is to perform an end-to-end analysis of banking data by integrating customer, account, loan, and transaction datasets. Using SQL, Python, Excel, and Power BI, the project transforms raw data into meaningful insights through data cleaning, analysis, visualization, KPI reporting, and interactive dashboards to support data-driven decision-making

Key Objectives

- Analyze customer demographics and account distribution.
- Monitor transaction performance and digital banking usage.
- Evaluate loan portfolio performance and lending trends.
- Develop interactive dashboards for executive decision-making.

Dataset Overview

Banking Analytics Dataset

The project utilizes four interconnected datasets representing different banking operations. These datasets are integrated using primary and foreign keys to support comprehensive analysis of customers, accounts, transactions, and loans.

Customers Table

Customer_ID	Customer_Name	Gender	Age	Email	City	Occupation	Join_Date	Credit_Score	KYC_Status	Phone_number	Age Group
CUST00001	Matthew Rana	Male	43	gautamsamra@example.org	Mumbai	Retired	Wednesday, October 26, 2022	598	Completed	1815908301	36-45
CUST00002	Udant Bhatia	Male	20	harshilguha@example.com	Hyderabad	Business	Thursday, April 29, 2021	596	Completed	Not Available	19-25
CUST00003	Dev Bhasin	Male	32	warda57@example.com	Chennai	Salaried	Sunday, May 2, 2021	845	Completed	7865797543	26-35
CUST00004	Jacob Vala	Male	20	eshan@example.org	Mumbai	Self Employed	Monday, April 12, 2021	623	Completed	2527601895	19-25
CUST00005	Urvi Bhattacharyya	Female	53	kamdarbhavna@example.org	Delhi	Retired	Friday, March 15, 2024	842	Completed	4974650752	46-55
CUST00006	Mitali Raman	Male	53	matthewbutala@example.org	Chennai	Retired	Thursday, February 2, 2023	655	Completed	1276842684	46-55
CUST00007	Vincent Kaul	Female	67	yutikabhagat@example.com	Chennai	Self Employed	Monday, December 25, 2023	735	Completed	7924402685	65+
CUST00008	Mitali Rastogi	Male	23	acharyamatthew@example.net	Hyderabad	Student	Tuesday, January 4, 2022	779	Completed	1760313721	19-25
CUST00009	Januja Bal	Male	50	tsule@example.org	Kolkata	Business	Sunday, May 30, 2021	800	Completed	3962459571	46-55
CUST00010	Harsh Oommen	Female	39	zarna52@example.com	Chennai	Retired	Tuesday, June 8, 2021	783	Completed	7280385280	36-45

Description: Contains customer demographic and profile information.

Key Columns:	Customer_ID (Primary Key)	City
	Customer_Name	Occupation
	Gender	Credit_Score
	Age	KYC_Status

Dataset Overview

Accounts Table

Account_ID	Customer_ID	Branch_ID	Account_Type	Open_Date	Balance	Account_Status
ACC00001	CUST00001	BR049	Savings	Saturday, December 3, 2022	₹ 26,209.79	Dormant
ACC00002	CUST00002	BR012	Savings	Thursday, July 22, 2021	₹ 67,604.26	Active
ACC00003	CUST00003	BR013	Savings	Monday, July 19, 2021	₹ 114,700.06	Active
ACC00004	CUST00004	BR036	Savings	Tuesday, July 13, 2021	₹ 196,687.69	Active
ACC00005	CUST00005	BR019	Savings	Tuesday, May 7, 2024	₹ 25,617.16	Active
ACC00006	CUST00006	BR007	Salary	Saturday, April 15, 2023	₹ 11,116.63	Active
ACC00007	CUST00007	BR034	Savings	Sunday, January 21, 2024	₹ 78,629.20	Active
ACC00008	CUST00008	BR045	Savings	Thursday, April 28, 2022	₹ 47,789.11	Active
ACC00009	CUST00009	BR032	Savings	Wednesday, August 11, 2021	₹ 137,292.63	Active
ACC00010	CUST00010	BR034	Salary	Friday, September 17, 2021	₹ 97,764.42	Active

Description: Stores account details associated with each customer.

Key Columns:

Account_ID (Primary Key)	Branch_ID
Customer_ID (Foreign Key)	Open_Date
Account_Type	Account_Status
Balance	

Dataset Overview

Transactions Table

Transaction_ID	Account_ID	Transaction_Date	Transaction_Time	Transaction_Type	Channel	Amount	Status	City
TXN0000001	ACC03517	Saturday, March 18, 2023	10:25:03 PM	Debit Card	POS	₹ 7,581.41	Success	Ahmedabad
TXN0000002	ACC03911	Sunday, August 27, 2023	5:46:33 AM	UPI	Mobile	₹ 4,011.31	Success	Pune
TXN0000003	ACC01915	Sunday, November 17, 2024	7:06:34 PM	IMPS	Mobile	₹ 6,339.35	Success	Chennai
TXN0000004	ACC02086	Friday, June 17, 2022	5:38:14 PM	UPI	Mobile	₹ 740.34	Success	Hyderabad
TXN0000005	ACC00754	Sunday, August 11, 2024	10:04:48 AM	ATM Withdrawal	ATM	₹ 1,406.27	Success	Mumbai
TXN0000006	ACC01542	Thursday, October 27, 2022	5:42:19 PM	Debit Card	POS	₹ 1,936.03	Success	Hyderabad
TXN0000007	ACC02904	Monday, November 17, 2025	5:27:52 AM	NEFT	Internet	₹ 1,484.35	Success	Kolkata
TXN0000008	ACC00042	Saturday, June 15, 2024	11:39:35 AM	ATM Withdrawal	ATM	₹ 2,007.17	Success	Hyderabad
TXN0000009	ACC00158	Sunday, February 26, 2023	10:24:09 PM	ATM Withdrawal	ATM	₹ 12,088.90	Success	Hyderabad
TXN0000010	ACC03837	Thursday, February 2, 2023	5:25:58 AM	Debit Card	POS	₹ 3,055.43	Success	Hyderabad

Description:

Contains daily banking transaction records used for transaction analysis.

Key Columns:

Transaction_ID (Primary Key)

Amount

Transaction_Time

Account_ID (Foreign Key)

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Transaction_ID (Primary Key)

Amount

Transaction_Time

Account_ID (Foreign Key)

City

Transaction_Date

Status

Transaction_Type

Channel

Dataset Overview

Loans Table

Loan_ID	Customer_ID	Account_ID	Loan_Type	Loan_Amount	Interest_Rate	Tenure_Months	Loan_Status	Approval_Date
LN00044	CUST03539	ACC03539	Personal	₹ 366,589	9.69%	36	Active	Thursday, July 7, 2022
LN00050	CUST01890	ACC01890	Personal	₹ 261,396	8.26%	36	Active	Monday, January 9, 2023
LN00071	CUST00478	ACC00478	Personal	₹ 363,156	8.30%	36	Active	Sunday, January 21, 2024
LN00086	CUST03044	ACC03044	Personal	₹ 496,600	11.64%	36	Active	Wednesday, November 8, 2023
LN00116	CUST01460	ACC01460	Personal	₹ 330,971	12.67%	36	Active	Monday, July 3, 2023
LN00128	CUST04762	ACC04762	Personal	₹ 576,807	7.00%	36	Active	Sunday, May 19, 2024
LN00152	CUST02236	ACC02236	Personal	₹ 653,896	14.23%	36	Active	Monday, September 11, 2023
LN00153	CUST02127	ACC02127	Personal	₹ 627,006	15.45%	36	Active	Thursday, July 21, 2022
LN00175	CUST02844	ACC02844	Personal	₹ 429,631	12.11%	36	Active	Thursday, February 17, 2022
LN00177	CUST01464	ACC01464	Personal	₹ 228,700	9.75%	36	Active	Sunday, March 16, 2025

Description: Stores loan-related information for customers.

<u>Key Columns:</u>	Loan_ID (Primary Key)	Interest_Rate
	Customer_ID (Foreign Key)	Tenure_month
	Loan_Type	Loan_Status
	Loan Amount	Approval_Date

Analysis Summary

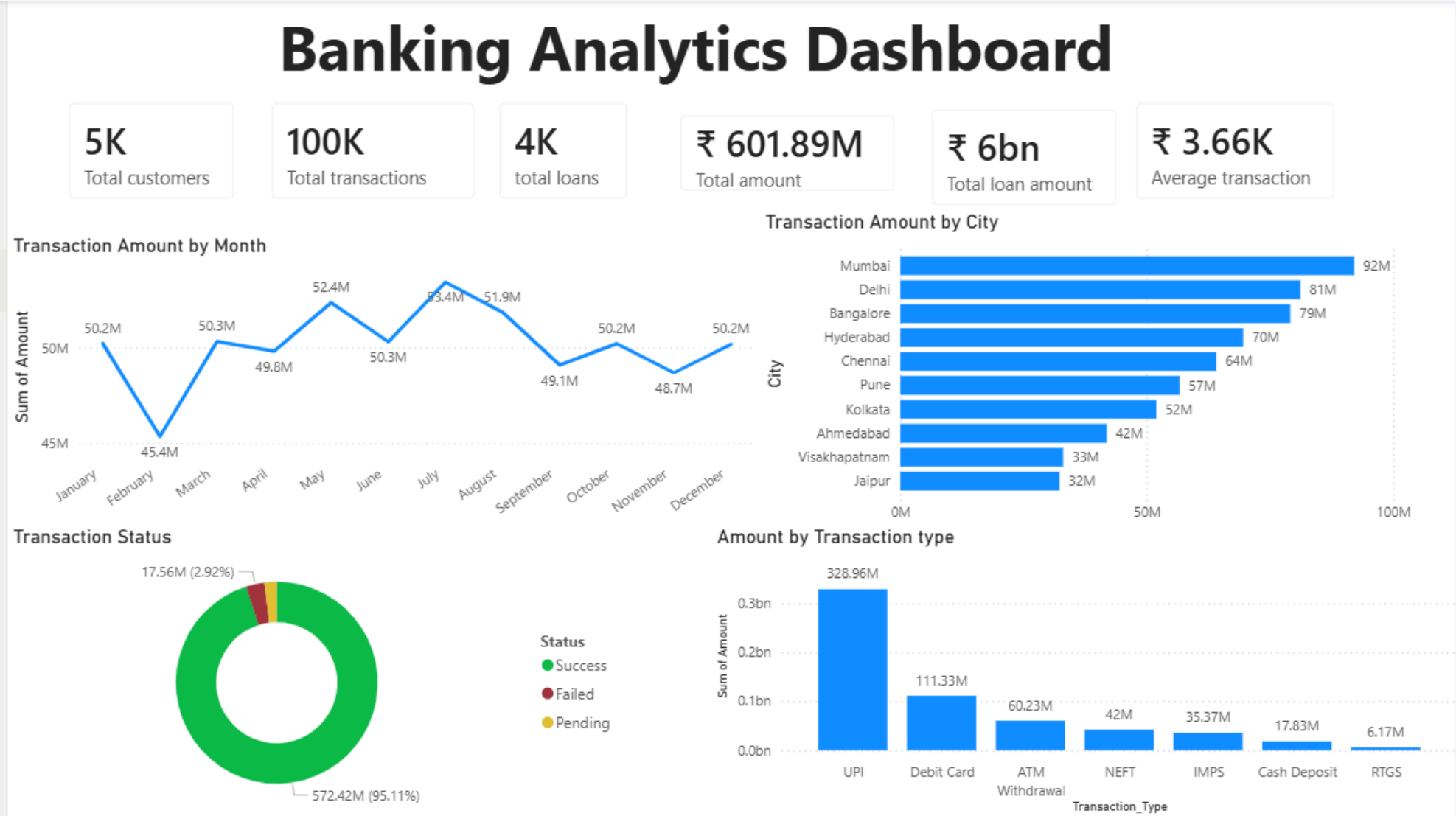
The banking dataset was analyzed using multiple analytical tools. Each tool was used for a specific purpose to transform raw data into meaningful business insights and interactive visualizations.

Tool Summary.

Tool	Analysis Performed	Key Outcome
Python	Data cleaning, exploratory data analysis (EDA), and visualizations	Identified trends, patterns, and prepared clean datasets.
SQL (MySQL)	Joins, aggregations, window functions, and business queries	Extracted meaningful insights from relational banking data.
Excel / Google Sheets	KPI calculations, formulas, Pivot Tables, and charts	Summarized business metrics for reporting.
Power BI	Interactive dashboard, KPI cards, charts	Built an executive dashboard for decision-making.

Analysis Summary

Power BI:

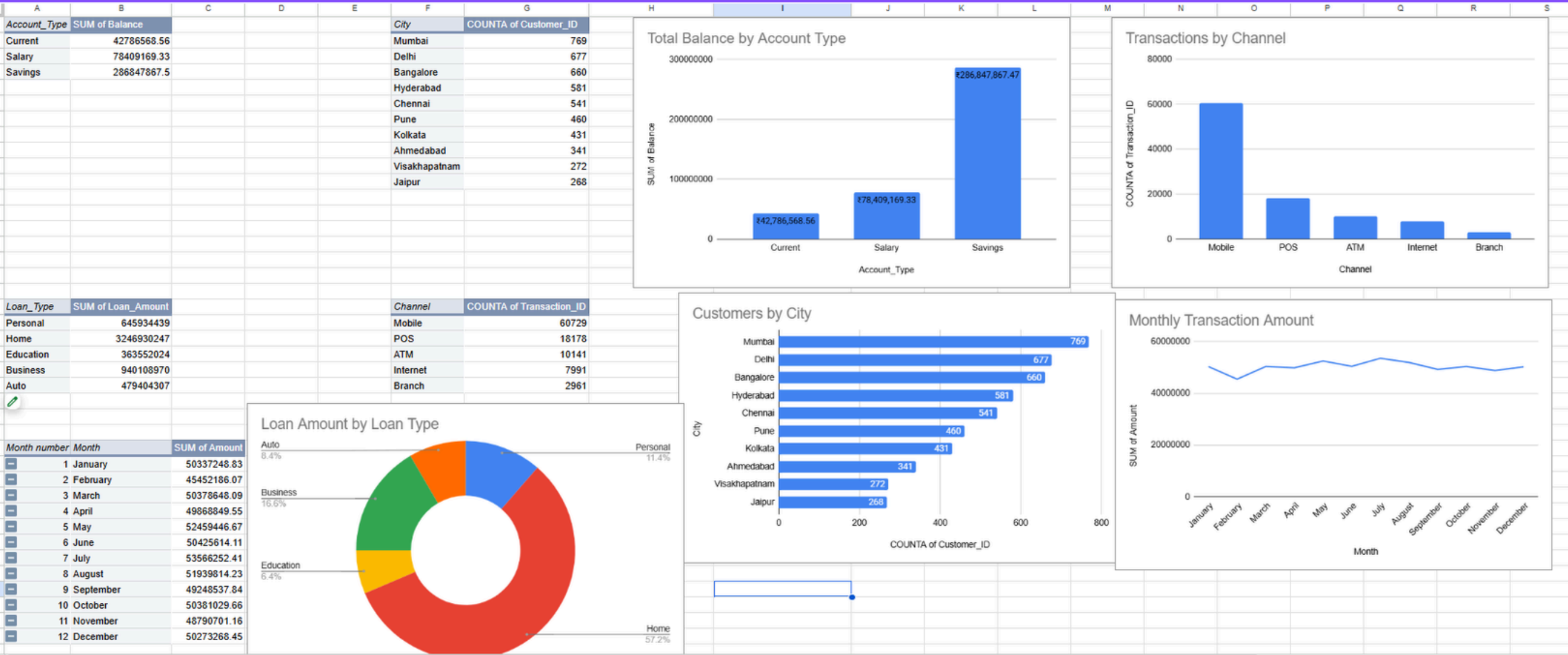


SQL:

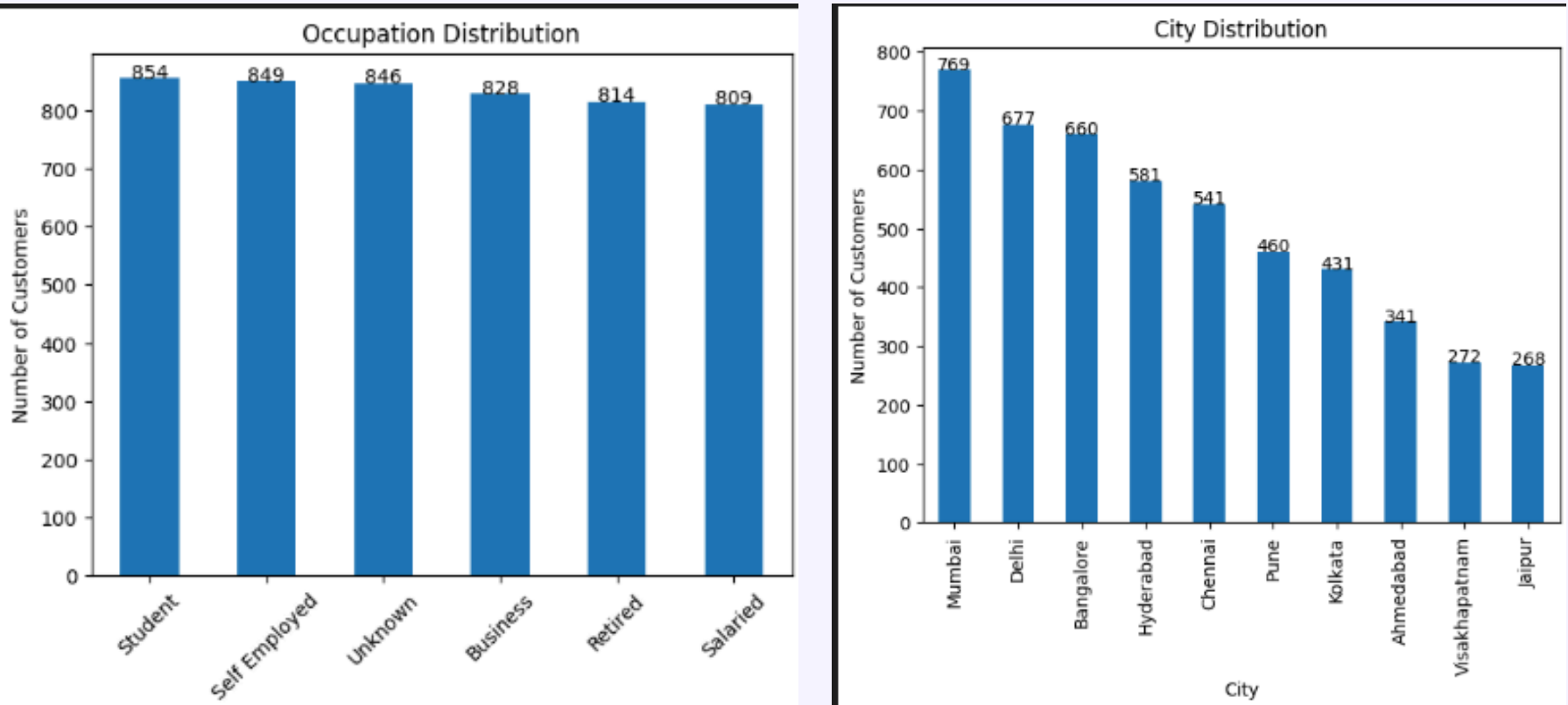
```
# Top 5 Customers by Total Account Balance
select c.Customer_Name,sum(a.Balance) as total_balance
from customers c
join accounts a
on c.i»¿Customer_ID=a.Customer_ID
group by c.Customer_Name
order by total_balance desc
limit 5;
```

Analysis Summary

Excel:



Python:



Insights:

- Savings accounts held the highest total balance, showing that they are the most preferred account type among customers for managing their funds.
- Digital banking channels were the most frequently used for transactions, indicating a strong shift toward convenient and technology-driven banking services.
- Home loans contributed the largest share of the bank's loan portfolio, making them one of the most important lending products.
- The analysis identified a group of customers with significantly higher account balances, highlighting an opportunity for the bank to offer premium banking services and personalized financial products.

Business Recommendations

Recommendations:

- Introduce premium banking services for high-value customers by offering personalized financial products, exclusive benefits, and dedicated relationship management.
- Continue investing in digital banking services to improve customer experience, as digital channels are the preferred mode of banking for most customers.
- Strengthen and promote high-performing loan products while maintaining effective credit risk assessment to ensure sustainable loan growth.
- Use data analytics regularly to monitor customer behavior and business performance, enabling faster decision-making and better strategic planning.

Conclusion:

This project provided me with practical experience in analyzing banking data using SQL, Python, Excel, and Power BI. Working with customer, account, loan, and transaction datasets helped me understand the complete data analytics process, from data cleaning and analysis to visualization and reporting.

By performing business analysis and building interactive dashboards, I was able to identify meaningful trends and generate insights that could support better business decisions. This project strengthened my technical skills and gave me hands-on experience in solving a real-world business problem using multiple analytics tools.

Overall, this project reflects my ability to work with data, extract valuable insights, and present them in a clear and meaningful way.

Banking Project Report

Thank You

Presented By : Vaddi pavan aditya manikanta